The cost of coalition withholding: resource dimension and coalition size

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Abstract

Owners contribute resource bundles to a common production technology. Payments must cover every coalition's stand-alone production and resist coordinated resource withholding, with side payments allowed inside the deviating coalition. We distinguish payment-only protection from protection that also compensates the coalition's outside production. For finite-valued monotone concave homogeneous technologies on d resources, the sharp worst-instance total-funding factors are $(d+1)/2$ and $1+c(d-1)$ respectively, where $c(2-e^{-1/c})=1$ and $c \simeq 0.5430633898$. For packing production of activity arity at most r, protecting coalitions of at most s owners admits a full-outside funding factor rs^{r-1}, independently of the owner and resource counts. A projective construction gives an unrestricted payment-only lower $K_3(s) \geq \sqrt{s/6}$, with a sharper explicit certificate asymptotic to $\sqrt{2\pi s}/(3\sqrt{e})$. It extends to $K_r(s) = \Omega_r(s^{(r-2)/(r-1)})$ for every fixed $r \geq 3$. All rules and deviations retain fixed actual owners and their full real report boxes. We give complete proofs, distinguish mixed and pure ownership, and supply exact finite certificate checks. The arity-three gap to $3s^2$ remains open. The quantitative claims are submitted for specialist correctness and priority assessment; the underlying monotonicity axioms, semivalues and hypergraph ingredients have established antecedents.

Keywords: coalition incentives; resource withholding; production games; subsidy; sparse activities; cost sharing.

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1 Introduction and relation to prior work

An efficient division can protect every coalition at one production profile and still fail when several owners withhold resources together. This paper studies the extra funding needed to choose payments consistently across all feasible reports. Owners may supply several inputs. Their identities, eligibility and endowment boxes remain fixed, and a protected coalition may include members whose reports do not change. Transfers inside the coalition make its total payment the relevant incentive quantity.

Two parameters organize the results. Resource dimension d measures the number of distinct inputs when every coalition is protected. Activity arity r instead bounds the inputs used by one production activity, while both the number of resources and owners may grow; s bounds only the size of coalitions receiving incentive protection. Static coverage remains unrestricted. A sponsor's factor κ bounds total payments by κ times realized production. The external supplement is therefore $(\kappa - 1)$ times production, not κ times production.

The paper joins the sharp dimension frontiers with a new lower-bound construction for bounded-coalition protection. The dimension results include mixed-owner upper formulas and a reduction from arbitrary finite-owner rules for the matching outside-production lower. The sparse upper is a binomial marginal construction using bounded-rank positive supports. The arity-three lower uses a constant-size productive seed, lawful symmetry of a fixed institution, and a ladder of one-owner restorations accompanied by unchanged protected members. The construction proves a funding obstruction for arbitrary payment functions, rather than only for a prescribed family of mechanisms. Appendix A gives the higher-arity and pure-owner versions in the same paper. The quantitative comparisons and proof dependencies are summarized in the separate specialist brief.

Demand monotonicity and absence-proofness. The aggregate-sum monotonicity used here has a close antecedent in strong group monotonicity of Moulin and Sprumont (2004). Their scalar integer-demand formulation permits passive members of the protected group. We do not claim that inequality as a new axiom. Their Proposition 2(ii), in the inspected 2004 author version, also assumes additivity, separability and exact balance; its impossibility does not directly apply to our unrestricted overfunded rules. The production class matters as well. Doğan (2013) studies absence-proofness with coalition outside worth in efficient subgame allocation schemes. That is a close conceptual comparison to full outside protection. Its inspected working-paper contract concerns binary absence and efficient subgames. Whether a published or related result implies a present quantitative bound is an explicit priority-review question.

Cross-monotonic sharing. Immorlica et al. (2008) give quantitative limitations of cross-monotonic cost and profit sharing. Their profit formulation has compatible overfunding and core signs. Its cross-monotonicity constrains each incumbent separately under participation changes, whereas our withholding condition constrains the sum for a coalition that can include a restoring owner. Their maximum-flow lower uses the individual constraints. It cannot be transferred solely from the theorem headline; an additional reduction would be needed. We make no claim that all related lower-bound techniques have been exhausted.

Static subsidies and restricted coalitions. The cost of stability of Bachrach et al. (2009) concerns subsidies making a single cooperative game stable, and already includes projective geometry and square-root phenomena. Neither ingredient alone is new here. Every packing profile in this paper already has an efficient static cover by dual prices; the obstruction is consistency across reports. In Meir et al. (2011), a coalition-size restriction removes static blocking constraints. Here

all coalition coverage constraints are retained, and the size restriction applies only to incentive comparisons. The parameters cannot be identified without changing the problem.

Positive supports and semivalues. The hierarchy of positive hypergraph representations developed by Feige et al. (2014), and the bounded-rank loss inequalities in Ezra et al. (2018), are antecedents for the sparse upper. Binomial marginal values and their relation to collusion are also established; see, for example, Casajus (2011). We include a direct proof of the exact upper used here, including full outside protection and mixed ownership, rather than attributing novelty to the probabilistic value itself. The central-binomial identity in the lower-bound ladder is a separate arithmetic ingredient.

Review status and scope. The proposed contribution is the collection of quantitative funding bounds under the stated contracts, especially the arbitrary-rule projective ladder. This draft is not a claim of verified historical priority. The review packet identifies original versions inspected, missing published-source comparisons, and additional cost-sharing leads. Internal AI-assisted audits and exact finite certificates do not replace specialist review of either correctness or novelty. No empirical pooling, power-reliability or opportunity-cost conclusion is included.

2 Model and funding frontiers

An institution fixes a finite set N of actual owners and d resource types. Owner i has an endowment $\bar{b}_i \in \mathbb{R}^d_+$ and may report any $b_i \in [0, \bar{b}_i]$ coordinatewise. Zero endowment coordinates encode ineligibility. These owners, coordinates, boxes and the public technology remain fixed throughout every comparison. Write $b_C = \sum_{i \in C} b_i$, $B = b_N$, and $\mathcal{B} = \prod_{i \in N} [0, \bar{b}_i]$. Different institutions may have different, arbitrarily large finite owner and resource counts.

The technology $F : \mathbb{R}^d_+ \to \mathbb{R}_+$ is finite-valued, monotone, concave and homogeneous of degree one, with $F(0) = 0$. These properties imply superadditivity:

$$F(x + y) = 2F((x + y)/2) \geq F(x) + F(y). \tag{1}$$

No closed-orthant continuity assumption is imposed on the general dimension class. In the sparse-production sections we specialize to a finite packing program

$$F(c) = \max\{\pi \cdot x : Ax \leq c,\ x \geq 0\}, \qquad A \geq 0, \quad \pi \geq 0. \tag{2}$$

Every positive-profit column consumes a positive amount of at least one resource. Zero-profit columns may be discarded. The optimum is finite and attained at finite capacities. Its *activity arity* is the maximum number of positive entries in a positive-profit column. Arity three permits many resource rows; it does not mean three resources in total.

A rule is an arbitrary map $y : \mathcal{B} \to \mathbb{R}^N_+$. It need not be continuous, anonymous, marginal, or linear. All requirements below concern the same rule at every real report profile. For $y_C(b) = \sum_{i \in C} y_i(b)$, *coalition coverage* is

$$y_C(b) \geq F(b_C) \qquad (C \subseteq N,\ b \in \mathcal{B}). \tag{3}$$

Coverage is required for *every* coalition, without a size restriction. A total-funding factor κ means

$$\sum_{i \in N} y_i(b) \leq \kappa F(B) \qquad (b \in \mathcal{B}). \tag{4}$$

Thus the external subsidy is at most $(\kappa - 1)F(B)$; κ itself is a factor on total payments. When $F(B) = 0$, nonnegativity and (4) force every payment to vanish. At a positive-production profile, an owner reporting zero is allowed to receive a positive payment. There is no deletion-marginal floor or separate dummy-zero axiom.

Definition 2.1 (Coalition withholding contracts)**.** Fix an integer $s \geq 1$. For every $C \subseteq N$ with $|C| \leq s$, let $\delta_i \in [0, b_i]$ for $i \in C$ and $\delta_i = 0$ outside C. *Transferable coalition withholding monotonicity* (TCWM) requires

$$y_C(b) - y_C(b - \delta) \geq 0. \tag{5}$$

Full outside-production protection (full OUT) requires

$$y_C(b) - y_C(b - \delta) \geq F(\delta_C). \tag{6}$$

Protected coalitions may contain unchanged members: $\delta_i = 0$ is allowed for any member of C. For unrestricted coalition protection, these inequalities are required for every $C \subseteq N$.

Both contracts permit side payments within the protected coalition. TCWM uses payment-only utility; full OUT additionally allows the coalition to produce with its pooled withheld resources. Neither is a model of arbitrary misreports, private opportunity costs, or an endogenous change in the owner roster. Full OUT implies TCWM. A lower bound for TCWM therefore also applies to full OUT, whereas a TCWM construction need not satisfy full OUT.

For an institution I, let $k_s(I)$ and $k_s^{\mathrm{OUT}}(I)$ be the infima of its feasible factors under these respective contracts and (3). Define

$$K_r(s) = \sup_{I:\, \mathrm{arity}(I) \leq r} k_s(I), \qquad K_r^{\mathrm{OUT}}(s) = \sup_{I:\, \mathrm{arity}(I) \leq r} k_s^{\mathrm{OUT}}(I). \tag{7}$$

The supremum ranges over all finite packing institutions, including their actual eligibility and endowment boxes. A uniform upper constructs one rule for each entire institution; a lower constructs an institution on which *every* rule has an expensive feasible report. The infimum convention does not assume optimal rules are attained. We use D_d and D_d^{OUT} for the corresponding worst-instance factors for general d-resource technologies when *every* coalition receives protection.

Remark 2.2 (Static feasibility versus cross-report incentives). For a packing technology, at each fixed profile an optimal nonnegative dual price w gives payments $w \cdot b_i$. Dual feasibility implies $w \cdot b_C \geq F(b_C)$, and their sum is $F(B)$. Thus each individual report profile admits an efficient coalition cover. The funding problem concerns selecting payments consistently across profiles. Neither the static core nor an integrality-gap calculation by itself proves our cross-report lower bounds.

3 The universal resource-dimension frontier

Here the technology may be any finite-valued function $F : \mathbb{R}_+^d \to \mathbb{R}_+$ that is coordinatewise nondecreasing, concave, and positively homogeneous of degree one. In particular, $F(0) = 0$. Continuity between different support faces is not assumed. The owners, their eligibility, and their finite coordinatewise endowment bounds are fixed throughout each mechanism. All real reports in those boxes, including zero reports, are in its domain.

Write D_d for the supremum, over these technologies and fixed finite owner models, of the infimal funding factor for nonnegative, coalition-covering TCWM payments. Define D_d^{OUT} in the same way

with full coalition outside-production protection. Write $Y_S(b) = \sum_{i \in S} y_i(b)$. Thus coverage means $Y_S(b) \geq F(\sum_{i \in S} b_i)$ for every coalition, and full OUT means

$$Y_S(b) - Y_S(b - w_S) \geq F\left(\sum_{i \in S} w_i\right), \qquad 0 \leq w_i \leq b_i \quad (i \in S). \tag{8}$$

TCWM replaces the right-hand side by zero. A coalition may contain members whose reports do not change. A funding factor K means $\sum_i y_i(b) \leq KF(\sum_i b_i)$ at every report, including those with zero production. These definitions do not presume attainment of a particular model's infimal factor.

Theorem 3.1 (Sharp universal dimension factors)**.** *For every integer $d \geq 1$,*

$$D_d = \frac{d+1}{2}, \qquad D_d^{\mathrm{OUT}} = 1 + c(d-1), \qquad c(2 - e^{-1/c}) = 1, \tag{9}$$

where c is the unique solution in $(1/2, 1)$ ($c \simeq 0.5430633898$). The upper bounds use explicit payment formulas for arbitrary fixed mixed owners. At a report with exactly $k \geq 1$ active resource coordinates they give the stronger respective factors $(k+1)/2$ and $1 + c(k-1)$; at the zero report they pay zero. The matching lower bounds already hold for $F(C) = \min_{1 \leq j \leq d} C_j$ on fixed finite endowment boxes, uniformly over the number of actual owners.

The sharpness is a worst-case statement across finite owner models. It does not assert the same lower bound for every technology or every fixed owner count. The lower bounds impose no anonymity, continuity, measurability, homogeneity, or price representation on the original mechanisms. The proof below constructs regular prices only as a consequence of assuming a uniform bound for all finite owner models.

3.1 Payment-only upper bound and a finite-owner lower bound

For aggregate report $C = \sum_i b_i$, let $I = \{j : C_j > 0\}$ and $k = |I|$. For $j \in I$ define the independent-uniform face average

$$A_j(C) = \int_{[0,1]^{I \setminus \{j\}}} F\left(C_j e_j + \sum_{\ell \in I \setminus \{j\}} u_\ell C_\ell e_\ell\right) du_{-j}, \qquad y_i(b) = \sum_{j \in I} \frac{b_{ij}}{C_j} A_j(C). \tag{10}$$

An integral over an empty set of coordinates is its integrand; all payments are zero when I is empty.

Proposition 3.2. *The rule (10) is nonnegative, covers every coalition, is TCWM at all real reports, and has total payment at most $(k+1)F(C)/2$ when $k \geq 1$.*

Proof. For coalition aggregate a and fixed outsider aggregate o, its coordinate-j payment is

$$\frac{a_j}{a_j + o_j} A_j(a + o).$$

Both nonnegative factors are nondecreasing under any coordinatewise increase of a. If a new coordinate becomes active, its own contribution is nonnegative. In each old face average, integrating over that newly active coordinate cannot lower the old integrand, whose coordinate was identically zero. This proves finite TCWM, including comparisons between different support faces.

For coverage, work in the k-dimensional active face. A finite concave function is locally Lipschitz and differentiable almost everywhere on its relative interior. At such a point x, its gradient $g(x)$ is nonnegative, and homogeneity gives $g(x) \cdot x = F(x)$. Moreover

$$g(x) \cdot z \geq F(z) \quad \text{for every nonnegative } z \text{ supported on this face.} \tag{11}$$

This includes boundary z: for $0 < t < 1$, apply the supporting inequality at x to $(1-t)x + tz$, and combine it with concavity along the same segment. After subtracting $F(x)$ and dividing by t, Euler's identity gives (11), without taking a boundary-continuity limit.

Let U_j be independent uniform random variables. For almost every fixed U_{-j}, one-dimensional absolute continuity on $[\varepsilon, 1]$ gives

$$C_j \int_\varepsilon^1 \partial_j F(u_j C_j, U_{-j} C_{-j})\, du_j = F(C_j, U_{-j} C_{-j}) - F(\varepsilon C_j, U_{-j} C_{-j}) \le F(C_j, U_{-j} C_{-j}).$$

The derivatives are nonnegative. Letting $\varepsilon \downarrow 0$ and using Tonelli therefore yields $\mathbb{E} g_j(U \odot C) \le A_j(C)/C_j$. Averaging (11) proves $\sum_j z_j A_j(C)/C_j \ge F(z)$. Taking z to be the aggregate of any actual coalition proves coverage.

Finally, two applications of concavity give

$$\begin{aligned} \sum_{j \in I} A_j(C) &\le \sum_{j \in I} F\left(C_j e_j + \tfrac{1}{2} \sum_{\ell \in I \setminus \{j\}} C_\ell e_\ell \right) \\ &\le k F\left(\frac{k+1}{2k} C \right) = \frac{k+1}{2} F(C). \end{aligned}$$

For $k = 1$ the same formula pays exactly $F(C)$; the empty face is immediate. □

Proposition 3.3 (A finite obstruction)**.** *Let $d \ge 2$ and $s \ge 2$ be integers. On $n = (d-1)s + 1$ fixed owners, each eligible for every resource and each with endowment box $[0,1]^d$, the technology $F(C) = \min_j C_j$ forces every nonnegative, coalition-covering TCWM rule to have funding factor at least*

$$1 + \frac{(s-1)((d-1)s - 1)}{2s^2 + (d-3)s - 1}. \tag{12}$$

Proof. Suppose the factor is κ, and put $\delta = \kappa - 1 \ge 0$. Average the rule over all permutations of the owner labels and resource labels. The fixed domain and technology are invariant under these finite groups. Nonnegativity, every coverage and TCWM inequality, and the funding bound are linear and hence are preserved. It suffices to consider this invariant rule; symmetry is not an assumption on the original mechanism.

At profile P_0, put s pure unit reporters in each resource group $1, \dots, d-2$, one pure unit reporter in each of the last two groups, and leave the other $s-1$ actual owners at zero reports. Let the respective per-owner payments be a, b, w, where $w \ge 0$. Selecting one owner from each resource group produces one, and grand production is one. Thus

$$(d-2)a + 2b \ge 1, \qquad (d-2)sa + 2b + (s-1)w \le \kappa.$$

Subtracting gives $(d-2)(s-1)a + (s-1)w \le \delta$, whence

$$2b \ge 1 - \frac{\delta}{s-1}. \tag{13}$$

When $d = 2$ the terms involving the absent first groups are zero.

At P_1, the $s-1$ zero reporters restore a pure unit of resource $d-1$. TCWM applied to these owners together with the original resource-$(d-1)$ owner shows that this group's new payment is at least $b + (s-1)w \ge b$. The first $d-1$ groups are now symmetric. If S is their union, then $Y_S(P_1) \ge (d-1)b$.

At P_2, every member of this same coalition S restores its report to one in each of resources $1, \dots, d-1$ and zero in resource d. These are feasible increases within the original unit boxes. Leave the last-resource owner unchanged. Write $m = (d-1)s$. All m members of S now have the same payment p, and write q for the last owner's payment. One member of S together with that owner produces one, so $p + q \geq 1$ while $mp + q \leq \kappa$. Consequently

$$(d-1)b \leq Y_S(P_1) \leq Y_S(P_2) = mp \leq \frac{m}{m-1}\delta.$$

Combining this with (13) and rearranging gives (12). All denominators are positive. All n owners remain present throughout; in particular, the proof never assumes that a zero reporter receives zero at a positive-output profile. □

Letting $s \to \infty$ in (12) and using Proposition 3.2 proves $D_d = (d+1)/2$ for $d \geq 2$. For $d = 1$, homogeneity makes F linear on its ray; linear payments and grand-coalition coverage give $D_1 = 1$.

3.2 The logarithmic rule for full outside production

Put $t_0 = 1 - e^{-1/c}$. The defining equation for c gives $c(1 + t_0) = 1$. The function $h(x) = x(2 - e^{-1/x})$ has

$$h'(x) = 2 - e^{-1/x}(1 + 1/x) > 1 \quad (x > 0),$$

and $h(1/2) < 1 < h(1)$, proving the asserted existence and uniqueness. Let U have density

$$f_U(u) = \frac{c}{1-u}\mathbf{1}_{(0,t_0)}(u). \tag{14}$$

It has mass $-c\log(1-t_0) = 1$ and mean $c[-t_0 - \log(1-t_0)] = c$. Use one common scalar within each face average:

$$H_j(C) = \mathbb{E}F\left(C_j e_j + U\sum_{\ell \neq j} C_\ell e_\ell\right), \qquad y_i(b) = \sum_{j: C_j > 0} \frac{b_{ij}}{C_j} H_j(C). \tag{15}$$

Absent-coordinate terms are zero. The independent-uniform and common-scalar averages are different rules.

Lemma 3.4 (Two scalar inequalities)**.** *For positive $x \in \mathbb{R}^k_{++}$, $k \geq 2$,*

$$\sum_{j=1}^{k} \frac{\mathbb{E}\min\{x_j, U\min_{\ell \neq j} x_\ell\}}{x_j} \geq 1. \tag{16}$$

If $H : [0,1] \to \mathbb{R}_+$ is nondecreasing and concave and $0 \leq t \leq t_0$, then, with derivatives taken almost everywhere,

$$\int_0^{t_0} [H(u) - uH'(u)]\, du = 2\int_0^{t_0} H(u)\, du - t_0 H(t_0) \geq t_0 H(t) - tH(1). \tag{17}$$

Proof. For (16), choose one smallest coordinate m, let S be the second-smallest coordinate, counting multiplicity, and put $t = m/S$. The smallest coordinate contributes

$$q(t) = \mathbb{E}\min\{1, U/t\} = \begin{cases} 1 - c - c(1-t)\log(1-t)/t, & 0 < t \leq t_0, \\ c/t, & t_0 \leq t \leq 1. \end{cases}$$

A second-smallest coordinate contributes ct, and all other terms are nonnegative. If $t \le t_0$, then $q(t) + ct - 1 = c(1-t)[-1 - \log(1-t)/t] \ge 0$. If $t \ge t_0$, then $q(t) + ct = c(t + t^{-1}) \ge 2c > 1$.

For (17), a possible downward jump at zero may be filled with the right limit. This does not change the integral and only strengthens the assertion when $t = 0$. A finite nondecreasing concave function is continuous from the left at one: concavity applied to a segment ending at one proves this by letting the segment parameter tend to one. After filling zero, therefore, H is continuous on $[0, 1]$. The integral identity follows by integration by parts from a positive endpoint and then passage to zero; boundedness gives $uH(u) \to 0$.

A continuous piecewise-linear nondecreasing concave function is a nonnegative constant plus a nonnegative linear function and a positive linear combination of hinges $\min\{u, s\}$, $0 < s \le 1$. For a constant A, the residual left side minus right side of the inequality is tA; for Bu it is $tB(1 - t_0)$. For the hinge with parameter s, the residual is respectively

$$\begin{cases} s(t-s), & s \le t, \\ (s-t)(t_0-s), & t < s < t_0, \\ t(s-t_0), & s \ge t_0. \end{cases}$$

Each is nonnegative. Linear interpolation on increasingly fine grids approximates a continuous concave function uniformly by such concave piecewise-linear functions. The integral identity, involving only values of H, passes the inequality to the limit. □

Proposition 3.5. *The rule* (15) *satisfies full OUT and coalition coverage at every real report, and its total payment is at most* $[1 + c(k-1)]F(C)$ *when* $k \ge 1$.

Proof. Nonnegativity is immediate. As in the proof of Proposition 3.2, the coalition's coordinate-j payment is the product of its nondecreasing share $a_j/(a_j + o_j)$ and the nondecreasing face value $H_j(a + o)$. The same formula also shows that an activated coordinate cannot lower any old contribution. Thus finite TCWM holds, including across support faces. Two Jensen inequalities give

$$\sum_j H_j(C) \le \sum_j F\left(C_j e_j + c \sum_{\ell \ne j} C_\ell e_\ell\right) \le [1 + c(k-1)]F(C).$$

First suppose F is smooth on the positive relative interior of every face. Fix a positive aggregate C in its active face, coalition holdings $0 \le a \le C$, and a nonnegative restoration direction z on this face. Write $\alpha_j = a_j/C_j$ and $v_j(u) = C_j e_j + u \sum_{\ell \ne j} C_\ell e_\ell$. Differentiating while both a and C increase in direction z gives

$$\begin{aligned} DY_S[z] &= \sum_j \{(1 - \alpha_j)L_j(z) + \alpha_j M_j(z)\}, \qquad (18) \\ L_j(z) &= \frac{H_j(C)}{C_j} z_j, \\ M_j(z) &= \mathbb{E}\left[U \nabla F(v_j(U)) \cdot z + (1-U)\partial_j F(v_j(U)) z_j\right]. \end{aligned}$$

The gradients are nonnegative and satisfy (11), so $L_j \ge 0$ and $M_j \ge cF(z)$. The expression is affine in each holding fraction α_j. It is therefore enough to prove $DY_S[z] \ge F(z)$ at every corner $\alpha \in \{0, 1\}^k$.

The result is immediate if $F(z) = 0$. For $k = 1$ the technology is linear on the active ray, and the assertion follows directly; assume $k \ge 2$. Initially suppose z is strictly positive on the active

face. If at least two holding fractions are one, the derivative is at least $2cF(z) \geq F(z)$. If all are zero, put $x_j = C_j/z_j$. Monotonicity and homogeneity imply

$$\frac{F(v_j(u))}{F(z)} \geq \min\{x_j, u \min_{\ell \neq j} x_\ell\}.$$

Equation (16) yields $\sum_j L_j(z) \geq F(z)$.

It remains to consider a corner with only $\alpha_j = 1$. Normalize $G(x) = F(z \odot x)/F(z)$, so $G(\mathbf{1}) = 1$. Common rescaling of $x = C/z$ leaves the normalized directional expressions in (18) unchanged. We may therefore arrange $\min_{\ell \neq j} x_\ell = 1$. Put $t = x_j$ and $V = G(x)$. Every face vector is at least ux, so the other coordinate terms satisfy

$$\sum_{\ell \neq j} \frac{L_\ell(z)}{F(z)} \geq cV \sum_{\ell \neq j} \frac{1}{x_\ell} \geq cV.$$

The first part of $M_j/F(z)$ is at least c. If $V \geq t_0$, their sum is at least $c(1 + t_0) = 1$. If $t \geq 1$, monotonicity gives $V \geq 1$, so this case is already covered. The only remaining case is $0 < t \leq V < t_0$.

Set $H(u) = G(te_j + u \sum_{\ell \neq j} x_\ell e_\ell)$. Then $H(t) \geq t$, $H(1) = V$, and Euler's identity gives

$$\partial_j G(te_j + ux_{-j}) = \frac{H(u) - uH'(u)}{t}.$$

Because $(1 - u)f_U(u) = c$ on $(0, t_0)$, the second part of $M_j/F(z)$ is at least

$$\frac{c}{t} \int_0^{t_0} [H(u) - uH'(u)]\, du \geq c(t_0 - V)$$

by (17). The singleton corner therefore contributes at least $c + c(t_0 - V) + cV = 1$. This completes every holding corner.

Differentiation under the expectation is justified locally by Euler's identity and nonnegative gradients:

$$C_j \partial_j F(v_j(u)) \leq F(C), \qquad uC_\ell \partial_\ell F(v_j(u)) \leq F(C) \quad (\ell \neq j).$$

These bounds dominate the differentiated integrands on every compact positive capacity set, including as $u \downarrow 0$. For directions with zero coordinates, apply the proved inequality to $z + \eta \mathbf{1}_I$ and let $\eta \downarrow 0$. The derivative expression is a finite linear function of its direction, while monotonicity gives $F(z + \eta \mathbf{1}_I) \geq F(z)$. No continuity of F across faces is used.

For a finite restoration z, take $a(s) = a(0) + sz$ and $C(s) = o + a(s)$, $0 \leq s \leq 1$. For every $s > 0$ its active face is the fixed face of $C(1)$. The derivative inequality integrates on $[\varepsilon, 1]$ to $Y_S(1) - Y_S(\varepsilon) \geq (1 - \varepsilon)F(z)$. Finite TCWM gives $Y_S(\varepsilon) \geq Y_S(0)$ even if the initial activation produces a payment jump. Letting $\varepsilon \downarrow 0$ proves full OUT for smooth F. A one-coordinate face is handled by its linear formula and the zero face is immediate.

Finally, choose a smooth probability density for Z supported in $[-1, 1]^d$ and let

$$F_\varepsilon(x) = \mathbb{E}_Z F(\mathrm{diag}(e^{\varepsilon Z_j})x).$$

This preserves nonnegativity, monotonicity, concavity, and degree-one homogeneity. Convolution in logarithmic coordinates makes it smooth on every positive relative face. Monotonicity and homogeneity give the pointwise bounds

$$e^{-\varepsilon} F(x) \leq F_\varepsilon(x) \leq e^{\varepsilon} F(x) \qquad \text{on every face.} \tag{19}$$

The smooth finite OUT inequalities thus pass to the limit, since (19) also bounds each face average and each nonnegative owner payment multiplicatively. This proves the claim for the entire stated class, including downward boundary jumps. Restoring a coalition's entire bundle from its zero report then proves coverage: its initial payment under (15) is zero. □

3.3 Why arbitrary finite-owner rules satisfy the matching OUT lower bound

The remaining argument concerns only the continuous technology $F_r(C) = \min_{1\le j\le r} C_j$. Its continuity imposes no restriction on the broader class used for the upper bound. We give the reduction in full to distinguish a fixed finite-owner impossibility from a restriction to price mechanisms.

Lemma 3.6 (Prices forced by bounded finite-owner models)**.** *Fix $r \ge 2$. Suppose that, for every N, the model with N fixed pure-resource owners of each of the r types, each with report interval $[0,1]$, admits a nonnegative, coalition-covering, full-OUT rule funded by κF_r. Then there are nonnegative degree-zero homogeneous prices $p(C)$ on $\mathbb{R}^r_{++}$, locally Lipschitz there, such that*

$$C \cdot p(C) \le \kappa F_r(C), \tag{20}$$

$$a \cdot p(C) \ge F_r(a) \qquad (0 \le a \le C), \tag{21}$$

$$a \cdot p(C) \ge (a-u) \cdot p(C-u) + F_r(u) \qquad (0 \le u \le a \le C). \tag{22}$$

When an aggregate has a zero coordinate, all its prices may be set to zero and these statements still hold at every nonnegative real profile. The prices can also be chosen symmetric under resource permutations.

Proof. For each N, average its rule over permutations within each resource type. At a binary profile with m_j active unit reporters of type j, write $\pi_j^{(N)}(m)$ for an active owner's payment. For integer $0 \le u \le a \le m$, coverage, budget, and the OUT comparison that turns u_j coalition members of type j completely off give

$$\begin{aligned} m \cdot \pi^{(N)}(m) &\le \kappa F_r(m), \\ a \cdot \pi^{(N)}(m) &\ge F_r(a), \\ a \cdot \pi^{(N)}(m) &\ge (a-u) \cdot \pi^{(N)}(m-u) + F_r(u). \end{aligned} \tag{23}$$

Inactive reporters may have positive payments. We have only dropped their nonnegative payments from the budget and from the right side of the last inequality. Set an unused coordinate $\pi_j(m)$ to zero when $m_j = 0$. For $m_j > 0$ the coordinate bound is $0 \le \pi_j^{(N)}(m) \le \kappa F_r(m)/m_j$, independently of N. A diagonal subsequence as $N \to \infty$ therefore gives one array $\pi(m)$ satisfying (23) for all integer count vectors. Every finite group of these constraints is available once N exceeds its largest count. The set $\mathcal{C}$ of arrays obeying these constraints and coordinate bounds is nonempty, compact in the countable product topology, and convex.

For a positive integer h let $(T_h\pi)(m) = \pi(hm)$. Multiplying all counts in (23) by h and dividing by h shows that T_h preserves $\mathcal{C}$. These linear maps commute. For the first L primes $\ell_1, \dots, \ell_L$, average $T_h\pi$ over $h = \prod_{j=1}^{L} \ell_j^{e_j}$, $0 \le e_j < L$. For a fixed prime ℓ_j, shifting its exponent changes this average by at most $\kappa F_r(m)/(Lm_v)$ in coordinate (m, v), by cancellation of the interior terms. A diagonal subsequential limit as $L \to \infty$ is fixed by every prime dilation, hence by every positive integer dilation.

For rational $C > 0$ define $p(C) = \pi(LC)$, where L clears the denominators. Integer-dilation invariance makes this independent of L and homogeneous under positive rational scaling. Clearing the denominators of C, a, u in (23) proves (20)–(22) for rational data.

Regularity follows from finite comparisons, rather than from any property of the original rules. For rational $C, C' > 0$, put $\lambda = \min_j C_j/C'_j$ and $v = C - \lambda C' \ge 0$. At aggregate $\lambda C'$, give a coalition all of resource j and no other resource; then restore v to that same coalition. TCWM, which is implied by OUT, and rational homogeneity yield

$$p_j(C') - p_j(C) \le \frac{v \cdot p(C)}{\lambda C'_j}. \tag{24}$$

On any compact box $[a,b]^r$ with $a>0$, the coordinate prices are bounded by $\kappa b/a$, the denominator is at least a^2/b, and

$$0 \le v_\ell \le \|C-C'\|_\infty(1+b/a).$$

Interchanging C, C' supplies the opposite inequality. Thus p is Lipschitz on the rational points of each such box and has a unique locally Lipschitz extension to the positive orthant. Its inequalities and homogeneity pass to real data by continuity. If coalition holdings or withheld amounts have zero coordinates, approximate retained holdings $a-u$, withdrawals u, and outsider holdings $C-a$ separately by nonnegative rational vectors, retaining their zero patterns, and reconstruct a, C by summation. This preserves $0 \le u \le a \le C$; both positive aggregate totals remain positive.

If $F_r(C)=0$, every subbundle of C has zero worth, so zero prices give coverage and budget on that face. If $C>0$ but $F_r(C-u)=0$, the right payment in (22) is zero, while coverage and monotonicity give $a\cdot p(C) \ge F_r(a) \ge F_r(u)$. These observations complete all boundary comparisons. Finally, average the constructed prices over the finite resource-permutation group. All the inequalities, homogeneity, and local Lipschitz continuity survive, and the resulting prices are resource-symmetric. □

Lemma 3.7 (Removing abundant coordinates)**.** *Under the hypothesis of Lemma 3.6, there is a symmetric, homogeneous, locally Lipschitz price rule for* $\min\{A,B\}$ *with nonnegative payments, coalition coverage, full OUT, finite funding factor* κ*, and the expenditure cap*

$$BP_2(A,B) \le DA \quad (B \ge A > 0), \qquad D = \frac{\kappa-1}{r-1}, \tag{25}$$

together with its symmetric counterpart.

Proof. At $(A,M,\dots,M)$, $0<A<M$, symmetry gives a short-resource price q and common long-resource price b. Coverage of a bundle with A units of every resource and the budget yield

$$q+(r-1)b \ge 1, \qquad Aq+(r-1)Mb \le \kappa A.$$

Subtracting A times the first inequality gives $Mb \le DAM/(M-A)$. At $(A,B,M,\dots,M)$ with $A \le B \le M$, restore a coalition owning all B units of the second resource to M units. TCWM gives

$$Bp_2(A,B,M,\dots,M) \le D\frac{AM}{M-A}. \tag{26}$$

For $r=2$, the left side is independent of M; letting $M\to\infty$ already proves the lemma using the prices from Lemma 3.6. Suppose henceforth $r \ge 3$.

For each positive rational pair (A,B), the first two prices at $(A,B,M,\dots,M)$ lie in $[0,\kappa]$. A common diagonal subsequence $M\to\infty$ gives rational pair prices P_1, P_2. They inherit symmetry and pair budget at most $\kappa\min\{A,B\}$. Equation (26) gives (25). For coalition pair holdings $(a,b) \le (A,B)$, add $\min\{a,b\}$ units of every extra resource to this coalition. The added payment is at most

$$\min\{a,b\}\sum_{j\ge 3} p_j(A,B,M,\dots,M) \le \frac{\kappa\min\{a,b\}\min\{A,B\}}{M},$$

which vanishes. Its worth is $\min\{a,b\}$, so pair coverage survives.

For OUT, it is essential to control the shifted extra totals in a deviation. For fixed $w \ge 0$, compare $z=(A,B,M,\dots,M)$ and $z'=(A,B,M-w,\dots,M-w)$, where $M>w$. Equation (24),

first in one orientation and then the other, gives for $j = 1, 2$

$$p_j(z') - p_j(z) \le \frac{w\kappa \min\{A, B\}}{M z_j}, \qquad (27)$$
$$p_j(z) - p_j(z') \le \frac{w\kappa \min\{A, B\}}{(M - w) z_j}.$$

Indeed, the forward residual consists only of w units in each extra coordinate; the reverse scaling is $(M - w)/M$ and its residual lies in the first two coordinates. The budget bounds both residual expenditures.

Fix rational pair holdings (a, b) and withdrawals $(u, v) \le (a, b)$, and put $w = \min\{u, v\}$. Give that coalition w units of every extra resource and withdraw all of them. Its full outside production is w. The high extra-coordinate payment vanishes as above, and its low retained extra holdings are zero. At the low pair $(A - u, B - v)$ with positive production, (27) shows that the prices with extra totals $M - w$ have the same subsequential limit as those with extra totals M. If that low pair has zero production, its entire payment is zero by budget. The limiting inequality is therefore

$$aP_1(A, B) + bP_2(A, B) \ge (a - u)P_1(A - u, B - v) + (b - v)P_2(A - u, B - v) + \min\{u, v\}.$$

Set prices to zero at zero-production pairs. We have obtained a rational pair rule satisfying all required finite inequalities and the cap.

This limit need not yet be homogeneous. Consider the compact convex set of rational pair-price arrays obeying these inequalities, symmetry, the budget, and (25). Each coordinate is bounded by $\kappa \min\{A, B\}/A$ or $\kappa \min\{A, B\}/B$. The operations $(T_h P)(A, B) = P(hA, hB)$ preserve the set. Repeat the prime-dilation averaging in Lemma 3.6; all its constraints are homogeneous and closed, so the resulting rational array is homogeneous and retains the cap. The finite comparison (24) now gives local Lipschitz continuity and extension to real positive pairs exactly as before. Coverage handles any zero-production low profile as in the last paragraph of that lemma. □

Lemma 3.8 (The two-resource obstruction)**.** *Every pair-price rule with the properties of Lemma 3.7 has $D \ge c$.*

Proof. For $A \ge B > 0$ write $t = B/A$, $p(t) = P_1(A, B)$ and $q(t) = P_2(A, B)$. These are well defined by homogeneity, locally Lipschitz for $0 < t \le 1$, and $p(t) \le Dt$. Budget gives $q(t) \le \kappa$, while symmetry at the positive balanced profile gives $q(1) = p(1) \le D$.

If a coalition owns all of the short resource and restores only that resource, TCWM implies that $z(t) = tq(t)$ is nondecreasing. To obtain the OUT constraint, start at aggregate (A, B) with coalition holdings $(0, B)$ and restore (h, h) to that same coalition. Its new holdings are $(h, B + h)$ and its new aggregate is $(A + h, B + h)$, with outsiders fixed. This is a feasible restoration from a zero holding, not a withdrawal from that zero holding. At almost every $t < 1$, differentiating its finite OUT inequality at $h = 0+$ gives

$$p(t) + q(t) + t(1 - t)q'(t) \ge 1.$$

Using the cap, this yields

$$\left[\frac{tq(t)}{1 - t}\right]' = \frac{q(t) + t(1 - t)q'(t)}{(1 - t)^2} \ge \frac{1 - Dt}{(1 - t)^2}.$$

Integrate from ε to t and send $\varepsilon \downarrow 0$. Boundedness of q makes the lower endpoint vanish, and hence

$$z(t) \geq (1-D)t - D(1-t)\log(1-t).$$

Since z is nondecreasing and $z(1) = q(1) \leq D$, the right side is at most D for every $0 < t < 1$. If $D = 0$, this is already a contradiction. For $D > 0$, its maximum occurs at $t = 1 - e^{-1/D}$ and equals $1 - D + De^{-1/D}$. Consequently $D(2 - e^{-1/D}) \geq 1$. The strict monotonicity of h proved above implies $D \geq c$. □

Completion of Theorem 3.1. The TCWM equality was proved after Proposition 3.3. Proposition 3.5 proves the OUT upper bound. For $d \geq 2$, suppose a factor $\kappa < 1+c(d-1)$ were feasible for every fixed finite pure-owner unit-box model of $F_d = \min_j C_j$. Lemmas 3.6–3.8 would give $(\kappa - 1)/(d-1) \geq c$, a contradiction. Therefore for each such κ at least one finite, fixed, bounded owner model has no κ-funded rule. This is a compactness contrapositive over finite models, not a mechanism on an infinite owner population. If the supremum of their infimal feasible factors were smaller than the claimed constant, choosing κ strictly between them would give just such a rule for every model. This also deals with possible nonattainment of individual infima. For $d = 1$, linear payments satisfy full OUT with factor one and coverage forces that factor. □

Remark 3.9. These dimension frontiers concern protection of total coalition payments, with the full value of withheld production added only in the OUT contract. They do not settle a sharp frontier for individual payment monotonicity. The displayed integrals are mathematical payment formulas; no efficient evaluation guarantee for an arbitrary value oracle is asserted. Intermediate outside-protection parameters and extra deletion-marginal floors are not included in the present theorem.

4 Sparse production: a uniform coalition-size upper bound

We first state the upper bound with which the new lower must be compared. The binomial semivalue and bounded-rank support inequalities are established ingredients (Feige et al., 2014; Ezra et al., 2018; Casajus, 2011). Their use here retains actual mixed owners, all-coalition coverage and the stronger full-OUT contract. The proof is included to expose the precise comparison class.

Theorem 4.1 (Sparse-production guarantee)**.** *For every finite packing institution of activity arity at most $r \geq 1$ and every integer $s \geq 1$, there is an all-profile rule satisfying coalition coverage and full OUT for every coalition of size at most s, with*

$$K_r(s) \leq K_r^{\mathrm{OUT}}(s) \leq rs^{r-1}. \tag{28}$$

For $p = 1/s$, one such rule is

$$m_i(p;b) = \mathbb{E}_{R \subseteq N\setminus\{i\}}[F(b_i + b_R) - F(b_R)], \qquad y_i(b) = p^{1-r} m_i(p;b), \tag{29}$$

where other owners are independently included in R with probability p. For rational data the rule is a finite rational computation. No polynomial-time implementation or incentive guarantee for a sampled estimator is asserted.

Lemma 4.2 (Owner support and thinning)**.** *Fix a report profile and let $v(T) = F(b_T)$. For $i \in R$, set $U_i(R) = v(R) - v(R \setminus \{i\})$. Then*

$$\sum_{i \in T} U_i(R) \geq v(T) \qquad (T \subseteq R), \tag{30}$$

$$\sum_{i \in R} U_i(R) \leq r v(R), \tag{31}$$

$$\mathbb{E} v(S \cap R_p) \geq p^r v(S), \tag{32}$$

where R_p independently retains every owner with probability p.

Proof. An optimal nonnegative dual price w at b_R satisfies $v(R) = w \cdot b_R$ and $v(R \setminus \{i\}) \leq w \cdot b_{R \setminus \{i\}}$. Thus $U_i(R) \geq w \cdot b_i$, and dual feasibility proves (30).

For a fixed S, choose an optimal production plan x. For every used activity e and each required row a, assign that row to an actual supplier $i \in S$ with probability $b_{ia}/(b_S)_a$, independently across required rows. The denominator is positive whenever $x_e > 0$ and $A_{ae} > 0$. Each assignment gives a nonnegative value piece, of weight $\pi_e x_e$ times its assignment probability, supported on at most r owners. Let $h(T)$ be the total weight of pieces whose suppliers all belong to $T \subseteq S$. For activity e, its retained fraction is

$$\prod_{a : A_{ae} > 0} \frac{(b_T)_a}{(b_S)_a},$$

which is at most the fraction in each individual required row. The retained plan is therefore feasible for b_T. Consequently $h(T) \leq v(T)$ and $h(S) = v(S)$. Applying this at $S = R$ gives

$$U_i(R) \leq h(R) - h(R \setminus \{i\});$$

summing counts each value piece at most r times and proves (31). Under independent thinning, a piece supported on $k \leq r$ owners survives with probability $p^k \geq p^r$. Hence $\mathbb{E} v(S \cap R_p) \geq \mathbb{E} h(S \cap R_p) \geq p^r v(S)$. This decomposes a production plan, not the strategic owners. □

Proof of Theorem 4.1. Let R_p now thin all owners. Conditioning on whether $i \in R_p$ gives

$$p \sum_{i \in S} m_i(p; b) = \mathbb{E} \sum_{i \in S \cap R_p} U_i(R_p) \geq \mathbb{E} v(S \cap R_p) \geq p^r v(S).$$

After multiplication by p^{-r}, this is coverage of every S, including $|S| > s$. Nonnegativity follows from monotonicity. Also

$$p \sum_i m_i(p; b) \leq r \mathbb{E} F(b_{R_p}) \leq r F(pB) = rpF(B),$$

where the second inequality is concavity. This proves the funding bound.

For $s \geq 2$, fix a protected coalition C and write $M_C = \sum_{i \in C} m_i$. Expanding the finite expectation yields

$$M_C(b) = \sum_{A \subseteq N} c_C(A) F(b_A),$$

with, for $0 < |A| < n$,

$$c_C(A) = p^{|A|-1}(1-p)^{n-|A|-1}(|A \cap C| - p|C|). \tag{33}$$

The endpoints are $c_C(\varnothing) = -|C|(1-p)^{n-1}$ and $c_C(N) = |C|p^{n-1}$. Since $p|C| \le 1$, every coefficient of a value affected by C's report is nonnegative. For a withdrawal δ supported on C, superadditivity gives $F(b_A) - F(b_A - \delta_A) \ge F(\delta_A)$. For $A \cap C = \varnothing$, both this difference and $F(\delta_A)$ vanish. Therefore

$$M_C(b) - M_C(b-\delta) \ge \sum_A c_C(A) F(\delta_A) = M_C(\delta).$$

The profile δ lies in the same endowment boxes, with the same owners and the same rule. Coverage at that profile gives $p^{1-r} M_C(\delta) \ge F(\delta_C)$, proving full OUT.

At $s = 1$, $p = 1$ and $m_i = F(B) - F(B - b_i)$. The second term is fixed under i's withdrawal, so superadditivity proves full OUT directly; the first two inequalities of Lemma 4.2 give coverage and factor r. If no positive-profit activity exists, the zero rule suffices. Empty coalitions and the empty owner set are immediate. □

5 A square-root lower bound for arity three

The following lower applies to arbitrary rules under the full report-domain contract. It does not impose a marginal formula, an efficient-price floor, anonymity, or a zero payment to inactive owners.

Theorem 5.1 (Projective cap exchange)**.** *For each integer $M \ge 1$, there is a unit-profit packing technology of arity three on $d = 2^{M+1} - 1$ resources, with $n = 2M + 1$ fixed fully eligible owners, for which every coalition-covering rule satisfying TCWM for coalitions of size at most $2M - 1$ has funding factor at least*

$$\max\left\{\frac{2}{3}Q_M,\ 1 + \frac{2(M-1)}{3M}Q_M\right\}, \qquad Q_M = \prod_{m=2}^{M} \frac{2m}{2m-1} = \frac{4^M}{2\binom{2M}{M}}. \tag{34}$$

Every owner's report domain is the full real unit cube. Consequently, for every integer $s \ge 1$,

$$\sqrt{s/6} \le K_3(s) \le K_3^{\mathrm{OUT}}(s) \le 3s^2. \tag{35}$$

5.1 Technology, legal symmetry and productive profiles

Let $V = \mathbb{F}_2^{M+1}$ have basis $v, e_1, \dots, e_M$. Resources are the nonzero vectors of V. For every unordered triple $\{a, b, a+b\}$ of distinct nonzero vectors, include one activity consuming one unit of each row and yielding one unit of output. Duplicate triples are identified. All activities have exactly three inputs.

There are actual owners $p, h_1, \ell_1, \dots, h_M, \ell_M$, each endowed with the all-ones vector. Every report in $[0,1]^d$ is legal for every owner. The product of owner permutations and invertible binary linear maps acts on this institution: the latter permute resources and activities, while owner permutations preserve the identical report domains.

Lemma 5.2 (Finite averaging)**.** *If a rule on this institution satisfies coverage, size-s TCWM (or full OUT) and factor κ, then its finite group average satisfies the same constraints and factor, and is equivariant. A lower on total payment of the average at a profile of production one is a lower on the original rule at some profile in that profile's finite symmetry orbit.*

Proof. For a group element g with owner permutation σ_g, define the conjugate rule by $y^g(b) = \sigma_g^{-1} y(gb)$. Resource and owner maps preserve coordinate order, report feasibility, coalition values, coalition sizes, and the production value of a withheld bundle. Each contract inequality therefore

holds for every conjugate. All constraints are linear in payment values, so their finite average also satisfies them. A change of variable in the group sum proves equivariance. Finally, its total payment at b is the average of original total payments at the legal profiles gb, which all have production $F(B)$. At least one is at least that average. This is a proof transformation of rules; no strategic comparison changes an owner. □

For $m = 1, \dots, M$, define profile C_m as follows. Owner p reports the unit vector in row v; each h_j reports the unit vector in row e_j; ℓ_j reports the unit vector in row $v + e_j$ if $j \le m$, and zero otherwise. Every unmentioned resource coordinate is zero. In particular, *dormant high owners still report* e_j when $j > m$.

Lemma 5.3 (Exact support and value)**.** *The only activities supported at C_m are the m triples $\{v, e_j, v + e_j\}$ with $j \le m$, and $F(C_m) = 1$.*

Proof. In the possible row set $\{v\} \cup \{e_j, v + e_j : 1 \le j \le M\}$, the sum of private rows from distinct indices has two nonzero e coordinates and leaves the set. The sum of the private pair at index j is v, and adding v exchanges those two rows. Thus every supported triple is precisely a complete active branch. Every branch consumes the reported unit of v, giving the upper bound one; one branch attains it. Equivalently, the full packing dual prices v and all zero-capacity rows at one and all other rows at zero. Supported columns contain v and unsupported columns contain a zero-capacity row, so this is a feasible dual of value one for the full technology. □

5.2 A constant seed and a fixed-owner payment ladder

Work with the averaged rule of Lemma 5.2. At C_1, the linear map exchanging v and e_1 and fixing e_j for $j > 1$, together with the owner swap (p, h_1), preserves the whole profile. So does the invertible map $v \mapsto v + e_1$, fixing every e_j, together with the swap (p, ℓ_1). These maps fix the dormant high reports. Hence the three productive owners receive a common payment a. Their coalition produces one, and coverage requires $3a \ge 1$. This uses static coverage for a three-member coalition, not a three-member incentive constraint.

At C_m, permuting active indices and independently replacing e_j by $e_j + v$ for active j preserves the entire profile after the matching owner swaps. These invertible maps fix v and all inactive high rows. They act transitively on the $2m$ active private owners. Let q_m be their common payment and put $T_m = 2mq_m$. The seed is

$$q_1 \ge \frac{1}{3}, \qquad T_1 \ge \frac{2}{3}. \tag{36}$$

The transition $C_{m-1} \to C_m$ restores only the actual owner ℓ_m. Protect the coalition consisting of ℓ_m and the $2m - 2$ old active private owners. Its cardinality is $2m - 1$, and every old member's report is unchanged. At the old profile its payment is at least T_{m-1}: $y_{\ell_m}(C_{m-1})$ is nonnegative, and is *not* assumed to be zero. At the new profile the protected sum is $(2m - 1)q_m$. Therefore TCWM gives

$$(2m - 1)q_m \ge 2(m - 1)q_{m-1}, \qquad T_m \ge \frac{2m}{2m - 1} T_{m-1}. \tag{37}$$

The unchanged new high owner h_m and the cap owner p are not in this protected coalition. No owner is split, cloned, added, or removed. Iterating (37) proves $T_M \ge 2Q_M/3$. Since production is one, nonnegativity and funding imply $\kappa \ge 2Q_M/3$.

Let $z = y_p(C_M)$. Terminal branch coverage gives $z + 2q_M \ge 1$, and total payment is at least $z + 2Mq_M$. Hence

$$\kappa \ge z + 2Mq_M \ge 1 + 2(M - 1)q_M \ge 1 + \frac{2(M - 1)}{3M} Q_M.$$

This remains valid even if $2q_M > 1$; it does not set $z = 1 - 2q_M$. For an asymmetric original rule, Lemma 5.2 locates the costly profile somewhere in the orbit, not necessarily at the displayed C_M. This proves the finite assertion of Theorem 5.1.

5.3 The uniform exponent and an optimized tail

The elementary estimate

$$Q_M \geq \frac{1}{2}\sqrt{3M+1} \tag{38}$$

follows by induction. The base $M = 1$ is equality; after squaring, the step reduces to $4(M+1)^2(3M+1) - (2M+1)^2(3M+4) = M \geq 0$. For $M = \lfloor (s+1)/2 \rfloor$, all incentive coalitions used have size at most s, and $M \geq s/2$. Thus $K_3(s) \geq 2Q_M/3 \geq \sqrt{3M+1}/3 \geq \sqrt{s/6}$. Combining this with Theorem 4.1 proves (35), including $s = 1$.

Corollary 5.4 (Optimized ladder certificate)**.** *Set $B_1 = 1$. For $s \geq 2$, let $h = \lceil s/2 \rceil$ and*

$$B_s = 1 + \frac{2(s-1)Q_h}{3h}\left(\frac{s-1}{s}\right)^{s-h}. \tag{39}$$

Then $K_3(s) \geq B_s$, and

$$\lim_{s\to\infty} \frac{B_s}{\sqrt{s}} = \frac{\sqrt{2\pi}}{3\sqrt{e}}. \tag{40}$$

The witness may use $2s+1$ fully eligible owners and $2^{s+1}-1$ resource rows. In particular $B_6 = 493/243 > 2$, with 13 owners and 127 rows. The optimization is of this ladder certificate, not of the unrestricted funding frontier.

Proof. Keep s fixed and permit arbitrary ladder length M. At stage m protect ℓ_m and any $k_m = \min\{s-1, 2m-2\}$ old active private owners. Whole pairs are unnecessary; all selected old owners have payments q_{m-1} and q_m at the respective profiles. TCWM and nonnegativity imply

$$(k_m+1)q_m \geq k_m q_{m-1}, \qquad q_M \geq \tfrac{1}{3}P_{M,s}, \qquad P_{M,s} = \prod_{m=2}^{M} \frac{k_m}{k_m+1}.$$

Terminal coverage therefore gives $\kappa \geq 1 + 2(M-1)P_{M,s}/3$. Up to h the uncapped recurrence gives $P_{h,s} = Q_h/h$; afterwards every factor is $\alpha = (s-1)/s$. For $M \geq h$ and $M > 1$, the consecutive ratio of $(M-1)P_{M,s}$ is $M\alpha/(M-1)$, which is greater than one for $M < s$, equal to one for $M = s$, and less than one for $M > s$. Before the cap the increment increases; its $M = 1$ value is zero and is handled separately. Thus the equal maxima occur at $M = s, s+1$, proving (39). The central-binomial asymptotic $Q_h \sim \sqrt{\pi h}/2$, together with $h \sim s/2$ and $\alpha^{s-h} \to e^{-1/2}$, proves (40). Finally $Q_3 = 8/5$ gives $B_6 = 493/243$. □

5.4 Exact certificate form and what computation verifies

The displayed argument has nonnegative rational multipliers. For fixed M, s, use the rows

$$3q_1 \geq 1, \qquad (k_m+1)q_m - k_m q_{m-1} \geq 0, \qquad z + 2q_M \geq 1, \qquad \kappa - z - 2Mq_M \geq 0.$$

Give the two final rows weight one. Starting with residual coefficient $2(M-1)$ at q_M, eliminate backwards: weight the row at m by the residual divided by $k_m + 1$, then replace the residual by that weight times k_m. Weight the seed by the final residual divided by three. All coefficients of

q_m and z cancel, leaving $\kappa \geq 1 + 2(M-1)P_{M,s}/3$. At $M = 1$ only the final rows are needed. A lifted certificate can expand averaging back into the original owner/profile variables; it retains the nonnegativity rows for dormant payments rather than replacing those payments by zero.

The accompanying standard-library checks verify small full projective technologies, the required linear maps, exact finite coefficient cancellation, and the optimized scalar tail. A selected original-variable lift additionally checks 34,440 inequalities cancelling 26,040 payment coordinates. These are reproducible finite certificates supporting the written argument. The universal group averaging and all-size geometry are conventional proofs, not statements proved merely by the finite tests.

6 Interpretation, scope and remaining questions

The two parameters answer different questions. The dimension frontier allows arbitrary technologies on d resources and protects every coalition. The sparse frontier limits each activity to r resources, allows d to grow, and protects coalitions of at most s owners while retaining coverage for all coalitions. A resource-dimension upper cannot be substituted for an owner-independent bound depending only on r, s.

At arity three, the square-root-to-quadratic gap remains open. The lower uses exponentially many resource rows. Pure ownership is possible, but requires more owners than the fully eligible construction; Appendix A keeps the counts separate. Neither the exponent nor its sharper constant claims minimal witness size. The simple binomial upper evaluates finitely many coalition production programs, potentially exponentially many. Efficient approximate evaluation requires separate incentive and coverage analysis.

The proposed contribution is the quantitative link between fixed-owner coalition withholding and funding, particularly the projective cap exchange that supplies a constant seed without a dummy-zero assumption. Projective geometry, positive hypergraph supports, semivalues, and coalition monotonicity are established ingredients. A specialist must assess whether a prior theorem, possibly under different terminology or normalization, already implies the present bounds. The accompanying priority matrix records inspected versions and unresolved original-source comparisons. Internal model reviews and exact certificates are not a historical-novelty certificate or human peer review.

A conditional infrastructure interpretation would give each owner genuine rights over compute inputs and firm power capacity, with physically feasible partial withdrawals and a sponsor who benefits from the stronger contract. Public telemetry does not establish those rights. The earlier bounded feasibility check supplied no compatible held-out outcome panel or sponsor surplus estimate. Moreover, a Gaussian reserve model is not automatically an arity-three packing technology. We therefore make no measured pooling, reliability, energy-saving or net-value claim here.

This manuscript concerns coordinated withholding. It does not settle the separate owner-count/resource-dimension gap for individual incentives in the companion truthful-production project. Optional floor parameters, structured production refinements and application measurement studies are not separate new claims of this draft. The immediate questions are correctness of the fixed-institution reductions, priority of the quantitative bounds, and whether these results form a useful unit of publication.

A Fixed arity and pure ownership

The projective construction of Theorem 5.1 extends to every fixed arity $r \geq 3$ by replacing binary triples with binary circuits of size r. This appendix gives the additional geometry and records

the exact finite bounds. It also gives a separate realization with one pure owner per resource. The resource count is exponential in the length of the construction in both realizations; the pure realization has exponentially many owners as well.

Throughout, payments are arbitrary nonnegative functions on the entire fixed real report domain. For a report profile $b = (b_i)_{i\in N}$, write $b_C = \sum_{i\in C} b_i$. The hypotheses used below are coalition coverage and a uniform funding bound,

$$\sum_{i\in C} y_i(b) \geq F(b_C) \quad (C \subseteq N), \qquad \sum_{i\in N} y_i(b) \leq \kappa F(b_N), \tag{41}$$

at every feasible profile, together with size-s coalition withholding monotonicity: if $|C| \leq s$, $b'_i \geq b_i$ coordinatewise for $i \in C$, and $b'_i = b_i$ for $i \notin C$, then

$$\sum_{i\in C} y_i(b') \geq \sum_{i\in C} y_i(b). \tag{42}$$

Members of C may have unchanged reports. Coverage in (41) is required for *every* coalition, including coalitions larger than s. No continuity, formula for payments, anonymity, or zero payment to a zero reporter is assumed. Since full OUT implies (42), every lower bound below also applies under full OUT.

A.1 The binary-circuit construction

Fix integers $k \geq 2$ and $M \geq 1$, and put $r = k + 1$. Let

$$V = \mathbb{F}_2^{1+(k-1)M}$$

have the specified basis

$$v, \qquad e_{j,\ell} \quad (1 \leq j \leq M,\ 1 \leq \ell \leq k-1).$$

Resource rows are indexed by $V \setminus \{0\}$, so their number is

$$d = 2^{1+(k-1)M} - 1. \tag{43}$$

A binary circuit means a nonempty set of distinct rows whose sum over $\mathbb{F}_2$ is zero and none of whose proper nonempty subsets has sum zero. The activities are all binary circuits of cardinality $k + 1$. Each activity has profit one and consumes one unit of each of its $k + 1$ rows. Thus, with $\mathcal{A}$ denoting the set of activities, the full technology is

$$F(c) = \max\left\{ \sum_{A\in\mathcal{A}} x_A : x_A \geq 0, \quad \sum_{\substack{A\in\mathcal{A}\\ u\in A}} x_A \leq c_u \text{ for every } u \in V \setminus \{0\} \right\}. \tag{44}$$

Every activity has arity exactly r, and every $g \in \mathrm{GL}(V)$ permutes the resources and the activities of this finite technology.

For the first realization take the fixed actual owner set

$$N = \{p\} \ \cup\ \{h_{j,\ell} : 1 \leq j \leq M,\ 1 \leq \ell \leq k-1\} \ \cup\ \{\ell_j : 1 \leq j \leq M\}.$$

There are $n = 1 + kM$ owners. Every owner has the all-ones endowment and the same fixed report cube $[0, 1]^d$. Let $\delta_u \in \mathbb{R}^d$ denote one unit of resource u, to distinguish real resource quantities from the binary vectors that index resources. Set

$$w_j = v + \sum_{\ell=1}^{k-1} e_{j,\ell} \quad \text{in } V.$$

Basis independence makes all displayed root, high and low rows nonzero and pairwise distinct. For $1 \le m \le M$, define a feasible profile C_m by

$$b_p = \delta_v, \qquad b_{h_{j,\ell}} = \delta_{e_{j,\ell}}, \qquad b_{\ell_j} = \begin{cases} \delta_{w_j}, & j \le m, \\ 0, & j > m. \end{cases} \tag{45}$$

All actual owners and all their report domains remain fixed as m changes. In particular, the high owners of inactive branches already report their basis rows.

Lemma A.1 (Supported activities)**.** *At C_m, the only activities whose rows all have positive capacity are*

$$A_j = \{v, e_{j,1}, \dots, e_{j,k-1}, w_j\}, \qquad 1 \le j \le m.$$

Consequently $F(b_N(C_m)) = 1$.

Proof. Consider a nonempty zero-sum subset of the larger set

$$S = \{v\} \cup \{e_{j,\ell} : j, \ell\} \cup \{w_j : 1 \le j \le M\}.$$

Let J be its set of selected low-row indices and put $t = |J|$. The basis coordinate $e_{j,\ell}$ occurs in only two members of S: the row $e_{j,\ell}$ itself and the row w_j. Cancellation therefore requires all $k-1$ high rows of every branch in J, and no high rows of any other branch. Cancellation of the v coordinate requires the root row precisely when t is odd. If $t = 0$, these conditions give the empty subset. Every nonempty zero-sum subset thus has cardinality

$$kt + (t \bmod 2). \tag{46}$$

For $k \ge 2$, this cardinality equals $k+1$ precisely when $t = 1$: if $t \ge 2$ it is at least $2k > k+1$. The corresponding set is A_j, and it is minimal because its k rows $v, e_{j,1}, \dots, e_{j,k-1}$ are independent and w_j is their sum. At C_m its low row has positive capacity exactly when $j \le m$, which proves the classification.

Running any one active branch at level one is feasible. Every supported activity consumes the root row, whose capacity is one, so production cannot exceed one. Equivalently, a dual certificate for the *full* technology assigns price one to v and to every zero-capacity row, and price zero to the other rows. A supported activity contains v; every other activity contains a zero-capacity row. Each dual constraint is therefore satisfied, and the dual objective is one. □

Lemma A.2 (Seed and private-owner orbits)**.** *After lawful finite averaging over resource automorphisms and owner permutations, the owners of the first branch at C_1 have a common payment $q_1 \ge 1/(k+1)$. At C_m, all km active private owners*

$$\{h_{j,\ell} : j \le m,\ 1 \le \ell \le k-1\} \cup \{\ell_j : j \le m\}$$

have a common payment q_m.

Proof. The finite averaging argument of Lemma 5.2 applies to $\mathrm{GL}(V) \times \mathfrak{S}_N$. A binary linear automorphism acts on a real report by *permuting its resource coordinates*; it does not take binary linear combinations of real capacities. These resource permutations preserve (44), the real report cubes and coordinatewise order. Owner permutations preserve the common fixed report domains, coalition sizes and the coverage and comparison inequalities. Consequently conjugating a rule and averaging preserves nonnegativity, (41), (42), and its funding factor. It suffices to prove the lower bound for this equivariant averaged rule.

At C_1, any permutation of the k basis vectors $v, e_{1,1}, \dots, e_{1,k-1}$ fixes their sum w_1 and fixes every high basis row of an inactive branch. Also the map

$$v \longmapsto w_1, \qquad e_{j,\ell} \longmapsto e_{j,\ell} \quad \text{for all } j, \ell$$

is invertible: the coefficient of v in the replacement for v is one. It exchanges v and w_1. These maps preserve the entire multiset of reported rows at C_1, including the inactive high reports and zero low reports. Combined with the matching owner permutations, they stabilize C_1 and act transitively on its $k+1$ first-branch owners. Their coalition can run A_1 at level one, so coverage gives $(k+1)q_1 \geq 1$. This use of coverage does not require $k+1 \leq s$.

At C_m, permute the active branch blocks, or permute the high basis vectors within one active branch, fixing v and the inactive basis vectors. Additionally, for each $j \leq m$ the basis replacement

$$e_{j,1} \longmapsto w_j$$

with every other basis vector fixed is invertible, since the coefficient of $e_{j,1}$ in w_j is one. It fixes v, exchanges $e_{j,1}$ with w_j, and fixes all other displayed rows except these two. Within-branch permutations and this exchange connect all k private rows of a branch; active branch permutations connect the branches. They preserve the entire profile, so the induced stabilizers are transitive on all km active private owners. Equivariance gives their common payment q_m. □

Theorem A.3 (Finite fixed-arity lower bound)**.** *Fix integers $k \geq 2$, $s \geq 2$ and $M \geq 1$. There is a finite institution of arity exactly $k+1$, with $1 + kM$ fixed fully eligible owners and $2^{1+(k-1)M} - 1$ resource rows, for which every payment rule satisfying (41)–(42) has funding factor at least*

$$L_{k,M,s} := 1 + \frac{k(M-1)}{k+1} P_{k,M,s}, \qquad P_{k,M,s} := \prod_{m=2}^{M} \frac{a_m}{a_m + 1}, \qquad a_m := \min\{s-1, k(m-1)\}. \tag{47}$$

Empty products are one. If $s \geq k(M-1)+1$, including $s = 1, M = 1$, the same construction gives

$$\kappa \geq \frac{k}{k+1} Q_{k,M}, \qquad \kappa \geq 1 + \frac{k(M-1)}{(k+1)M} Q_{k,M}, \qquad Q_{k,M} := \prod_{m=2}^{M} \frac{km}{k(m-1)+1}. \tag{48}$$

Proof. Use (45) and the averaged rule from Lemma A.2. In the transition from C_{m-1} to C_m, only the same actual owner ℓ_m changes its report, from 0 to δ_{w_m}. Select any a_m of the $k(m-1)$ old active private owners and protect them together with ℓ_m. This fixed coalition has $a_m + 1 \leq s$ members. All selected old members keep their reports unchanged. At the old profile their total payment is $a_m q_{m-1}$, and the restoring owner's old payment is nonnegative. At the new profile all selected members, including the restorer, receive q_m. Hence (42) implies the exact recurrence

$$(a_m + 1) q_m \geq a_m q_{m-1}, \qquad 2 \leq m \leq M. \tag{49}$$

The selected old owners need not comprise whole branches. Iterating (49) from $(k+1)q_1 \geq 1$ gives

$$q_M \geq \frac{P_{k,M,s}}{k+1}. \tag{50}$$

Write z for the root owner's payment at C_M. Coverage for the actual owners of any complete branch and the budget at C_M give respectively

$$z + k q_M \geq 1, \qquad \kappa \geq z + kM q_M.$$

The second inequality uses Lemma A.1 and discards every other owner's payment only by nonnegativity. Combining them yields $\kappa \geq 1 + k(M-1)q_M$, proving (47). This includes $M = 1$, when terminal coverage alone gives $\kappa \geq 1$.

If $s \geq k(M-1)+1$, protect *all* old active private owners at every transition. The recurrence becomes

$$[k(m-1)+1]q_m \geq k(m-1)q_{m-1}.$$

Writing $T_m = kmq_m$ gives

$$T_1 \geq \frac{k}{k+1}, \qquad T_m \geq \frac{km}{k(m-1)+1} T_{m-1}.$$

Thus $T_M \geq kQ_{k,M}/(k+1)$, and the budget proves the first bound in (48). Since $q_M \geq Q_{k,M}/[(k+1)M]$, terminal branch coverage proves the second. When $M = 1$ there are no comparisons, so this argument also applies at $s = 1$.

All comparisons just used occur inside the one original fixed institution. Symmetry is a proof reduction, not a restriction on admissible payment rules. If the original rule is asymmetric, an expensive profile may be an image of C_M under the finite symmetry group, rather than C_M itself. The same lower bound on its worst-case funding factor follows. □

The finite bound also has a direct nonnegative rational certificate. Put $c = k(M-1)$ and $P = P_{k,M,s}$. Give the terminal coverage inequality $z + kq_M \geq 1$ and the budget inequality $\kappa - z - kMq_M \geq 0$ weight one each. Give the seed inequality $(k+1)q_1 \geq 1$ weight $cP/(k+1)$, and give recurrence m weight

$$\lambda_m = \frac{c}{a_m + 1} \prod_{j=m+1}^{M} \frac{a_j}{a_j + 1}. \tag{51}$$

For $M \geq 2$, the coefficient of q_M from the final two inequalities is $-c$, canceled by $\lambda_M(a_M + 1)$. For $2 \leq m < M$, $\lambda_m(a_m + 1) = \lambda_{m+1}a_{m+1}$, and $\lambda_2 a_2 = cP$ cancels the seed term. The z coefficients also cancel, leaving exactly $\kappa \geq 1 + cP/(k+1)$. At $M = 1$, $c = 0$ and terminal coverage plus budget already give the conclusion. These are consequences for arbitrary payment variables; no marginal-payment representation is used.

A.2 Length optimization and the fixed-arity exponent

Corollary A.4 (Optimized length within the construction)**.** *For fixed integers $k \geq 2$ and $s \geq 2$, set*

$$h = \left\lceil \frac{s}{k} \right\rceil, \qquad \rho = \frac{s-1}{s}.$$

The largest value of $L_{k,M,s}$ over positive integer lengths M is attained exactly at $M = s$ and $M = s+1$, with common value

$$B_{k,s} = 1 + \frac{k(s-1)}{(k+1)h} Q_{k,h} \rho^{\,s-h}. \tag{52}$$

The length $M = s$ realizes this bound with $n = 1 + ks$ fully eligible owners and $d = 2^{1+(k-1)s} - 1$ rows. For fixed k,

$$B_{k,s} - 1 \sim \frac{k}{k+1} \Gamma\left(1 + \frac{1}{k}\right) k^{1/k} \exp\left(-\frac{k-1}{k}\right) s^{(k-1)/k} \qquad (s \longrightarrow \infty). \tag{53}$$

This optimizes the displayed certificate over its length; it does not identify the optimal funding factor among all mechanisms.

Proof. The cutoff satisfies $k(h-1) \le s-1$ and $kh \ge s$. Through $m = h$ the recurrence is untruncated, and for $m \ge h+1$ its factor is ρ. Telescoping the factors $(m-1)/m$ therefore gives

$$P_{k,h,s} = \frac{Q_{k,h}}{h}, \qquad P_{k,M,s} = \frac{Q_{k,h}}{h}\rho^{M-h} \quad (M \ge h).$$

Let $I_M = (M-1)P_{k,M,s}$. We have $I_1 = 0 < I_2$. If $1 < M < h$,

$$\frac{I_{M+1}}{I_M} = \frac{kM^2}{(M-1)(kM+1)} > 1,$$

because the numerator minus the denominator is $(k-1)M + 1 > 0$. If $M \ge h$ and $M > 1$, the ratio is

$$\frac{I_{M+1}}{I_M} = \frac{M\rho}{M-1},$$

which exceeds one for $M < s$, equals one for $M = s$, and is less than one for $M > s$. As $h \le s$, this proves the two asserted maximizers, including the boundary case $h = 1$. Substituting $M = s$ proves (52).

The gamma functional equation gives the exact identity

$$Q_{k,h} = \frac{\Gamma(h+1)\Gamma(1+1/k)}{\Gamma(h+1/k)}.$$

Stirling's formula yields $Q_{k,h} \sim \Gamma(1+1/k)h^{(k-1)/k}$. Moreover $h \sim s/k$ and $\rho^{s-h} \longrightarrow \exp(-(k-1)/k)$. Inserting these limits into (52) proves (53). □

Corollary A.5 (An all-s bound)**.** *For every fixed integer $r \ge 3$ and every integer $s \ge 1$, writing $k = r-1$, the unrestricted worst-case optimal funding factor satisfies*

$$K_r(s) \ge \frac{k}{k+1}\left(\frac{s}{2k}\right)^{(k-1)/k}. \tag{54}$$

In particular, $K_r(s) = \Omega_r(s^{(r-2)/(r-1)})$.

Proof. Put $\beta = (k-1)/k$. For $m \ge 2$, concavity of x^β gives

$$\frac{km}{k(m-1)+1} = \frac{m}{m-\beta} \ge 1 + \frac{\beta}{m} \ge \left(1 + \frac{1}{m}\right)^\beta.$$

Consequently $Q_{k,M} \ge ((M+1)/2)^\beta$, also at $M = 1$. Choose $M = \lceil s/k \rceil$. Then $k(M-1)+1 \le s$ and $M \ge s/k$, so the first bound in (48) implies (54). The finite institution is admissible for the worst-case class defining $K_r(s)$. □

A.3 A separate realization with pure owners

Corollary A.6 (Fixed pure-owner realization)**.** *All the lower bounds above also hold for institutions with one fixed pure unit owner per resource row. For the length-M instance, this realization has*

$$n = d = 2^{1+(k-1)M} - 1, \tag{55}$$

and every owner's fixed report domain is the full real interval along its own resource coordinate. It does not have the fully eligible realization's owner count $1 + kM$.

Proof. Keep the technology (44). For each row $u \in V \setminus \{0\}$ introduce the actual owner i_u with fixed endowment δ_u and report domain

$$\{t_u \delta_u : 0 \le t_u \le 1\}.$$

All these owners exist at every profile. A resource automorphism g acts *jointly* on resource rows and owner labels, sending owner i_u to owner $i_{g(u)}$. On scalar reports the action is

$$(g \cdot t)_{g(u)} = t_u.$$

This preserves the fixed institution: the transformed owner still reports only its own resource. The associated owner bijection takes a coalition to another coalition of the same size, maps its total resource report by the same technology automorphism, preserves coordinatewise restoration, and leaves all outsiders unchanged in a comparison. Conjugation therefore preserves every coverage, funding and incentive inequality. Averaging over this finite joint $\mathrm{GL}(V)$ action is lawful. No independent owner permutation is required or used.

At the pure-owner version of C_m, set $t_u = 1$ exactly for

$$S_m = \{v\} \cup \{e_{j,\ell} : j, \ell\} \cup \{w_j : j \le m\},$$

and set $t_u = 0$ for every other row. Its aggregate capacities coincide with those of (45), so Lemma A.1, including the full-technology dual, applies without change. Each seed or private stabilizer described in Lemma A.2 preserves S_m as a set. Being a permutation of all nonzero rows, it also preserves its complement. Its joint action thus fixes the entire report profile, including every additional zero-report owner. The same seed and private orbits give $(k+1)q_1 \ge 1$ and a common payment q_m for the km active private owners.

In the transition from C_{m-1} to C_m, only the same actual pure owner i_{w_m} restores its report from zero to one. Protect this owner together with the specified old active pure owners. Their labels, eligibility and reports remain fixed throughout the comparison. Equation (49) follows exactly as before, by discarding only the restorer's nonnegative old payment. Terminal branch coverage involves the $k+1$ actual pure owners of that branch. All omitted owners, including additional zero reporters, have nonnegative payments, so the terminal budget argument is unchanged. This proves the finite bounds and all their stated consequences with the counts (55). □

For example, $k = 2$ specializes to the arity-three construction. Its optimized length-s bound has coefficient $\sqrt{2\pi}/(3\sqrt{e})$ in front of $\sqrt{s}$, and $B_{2,6} = 493/243$. The length-six witness uses 127 resource rows in either realization, with 13 fully eligible owners or 127 pure owners. Neither the fixed-arity extension nor the change of ownership realization gives a bound with a fixed resource dimension.

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